Python for Data Science 2

Lab 7- SUSY at LHC

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- To understand the most fundamental phenomena in nature, the Physicists application of the scientific method:

  1. Build Models

  - The **Standard Model of HEP** describes the building blocks of matter and their interactions.
    - SM requires the existence of a set of fundamental particles with very specific properties.
      - Tested and validated through experiments, including some of the most precise measurements ever.
    - Yet, has **failures**, e.g. no Dark Matter. Not compatible with Gravity.
    - And, has **inconsistencies**: e.g. both SM predicts the Higgs mass to be big and requires it to be small. (Known as the Hierarchy problem…
  - Models are **expressed mathematically**
    - Ideally build on a minimal set of rules (principles/laws)
    - SM Mathematical Framework: Quantum Field Theory- Way beyond this class.
      - Renormalization: encapsulating unknown (effects at high energy) in measurable parameters.
      - SM is a Model with **19 parameters**. Masses of particles. Strengths of forces. etc…
        - None are fundamental like speed of light or Planck's constant.

  2. **Build new model that tackle inconsistencies**.

  3. **Create experiments (target weaknesses in the model)**

Our Mathematical Models

Classical:
Calculus (Infinitesimal)
Object (a particle) described by x, y, z, α,β,γ and their derivatives (ie momentum).

Atomic scales

Quantum Mechanics:
Probabilistic
A particle described by a complex wave-function P(x, y, z, α,β,γ + derv).

Relativity explains Electromagnetic Unification. EM fields inherently relativistic.

General Relativity:
Tensor Calculous
Gravity = Curvature of Space Time.
Gravity weak, so curvature ignorable on small scales.

Relativistic QM Requires Creation of Particles... leads to Fields

Classical Field Theory:
eg Electrodynamics
Interaction of particles with a dynamic field A(x, y, z).

Relativistic Particles

Quantum Field Theory:
eg: QED, Standard Model
Particles = Fields.
Gauge Symmetries = the classical dynamic Fields = Particles = Electroweak
Calculate Probability of interaction.

Building the SM

Field Theory + Quantum Mechanics + Relativity = Quantum Field Theory

Ingredients

1. Leptons + Quarks

FERMIONS matter constituents spin = 1/2, 3/2, 5/2, ...

Leptons spin = 1/2		
Flavor	Mass GeV/c^2	Electric charge
ν_e electron neutrino	$<1\times10^{-8}$	0
e electron	0.000511	−1
ν_μ muon neutrino	<0.0002	0
μ muon	0.106	−1
ν_τ tau neutrino	<0.02	0
τ tau	1.7771	−1

Quarks spin = 1/2		
Flavor	Approx. Mass GeV/c^2	Electric charge
u up	0.003	2/3
d down	0.006	−1/3
c charm	1.3	2/3
s strange	0.1	−1/3
t top	175	2/3
[illegible]	4.3	−1/3

2. Three Local Gauge Symmetries

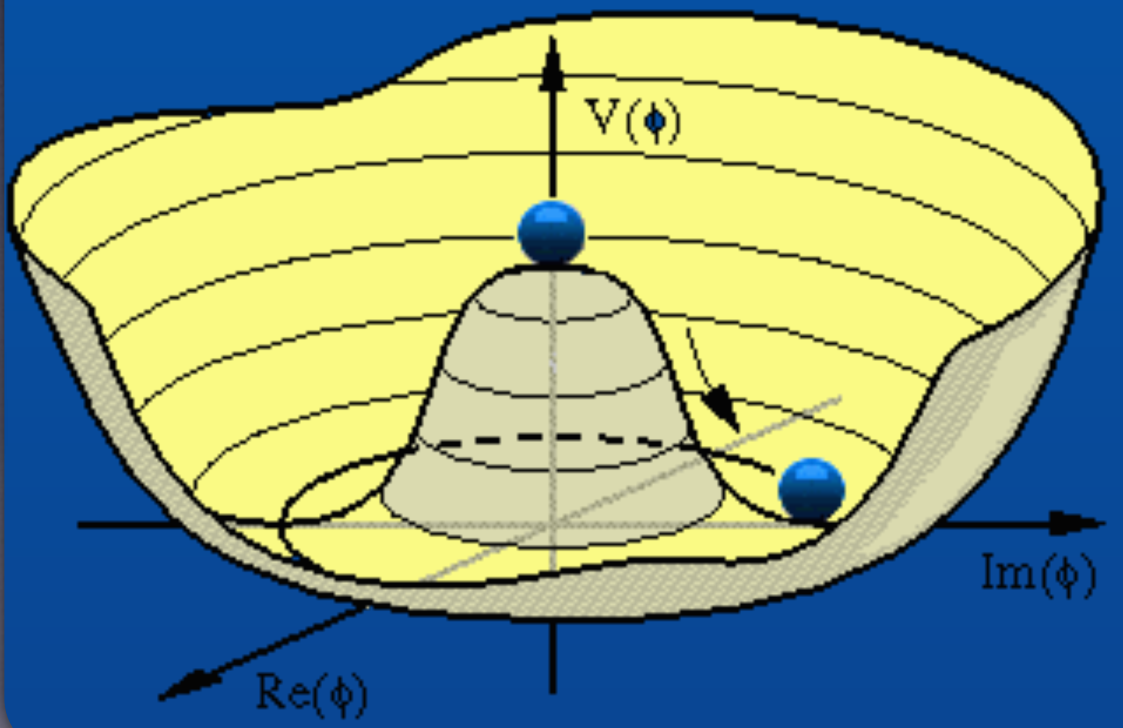
- Each Symmetry implies existence of a new set of boson (spin 1) fields
- These bosons "carry" the Forces

Unified Electroweak spin = 1		
Name	Mass GeV/c^2	Electric charge
γ photon	0	0
W^-	80.4	−1
W^+	80.4	+1
Z^0	91.187	0

Strong (color) spin = 1		
Name	Mass GeV/c^2	Electric charge
g gluon	0	0

3. Higgs Mechanism

- Every particle interacts with a scalar (spin 0) field
- This field has non-zero Vacuum Expectation Value
  - Breaks symmetry between E&M and Weak Forces
  - Gives masses to all particles w/o breaking Gauge Symmetries

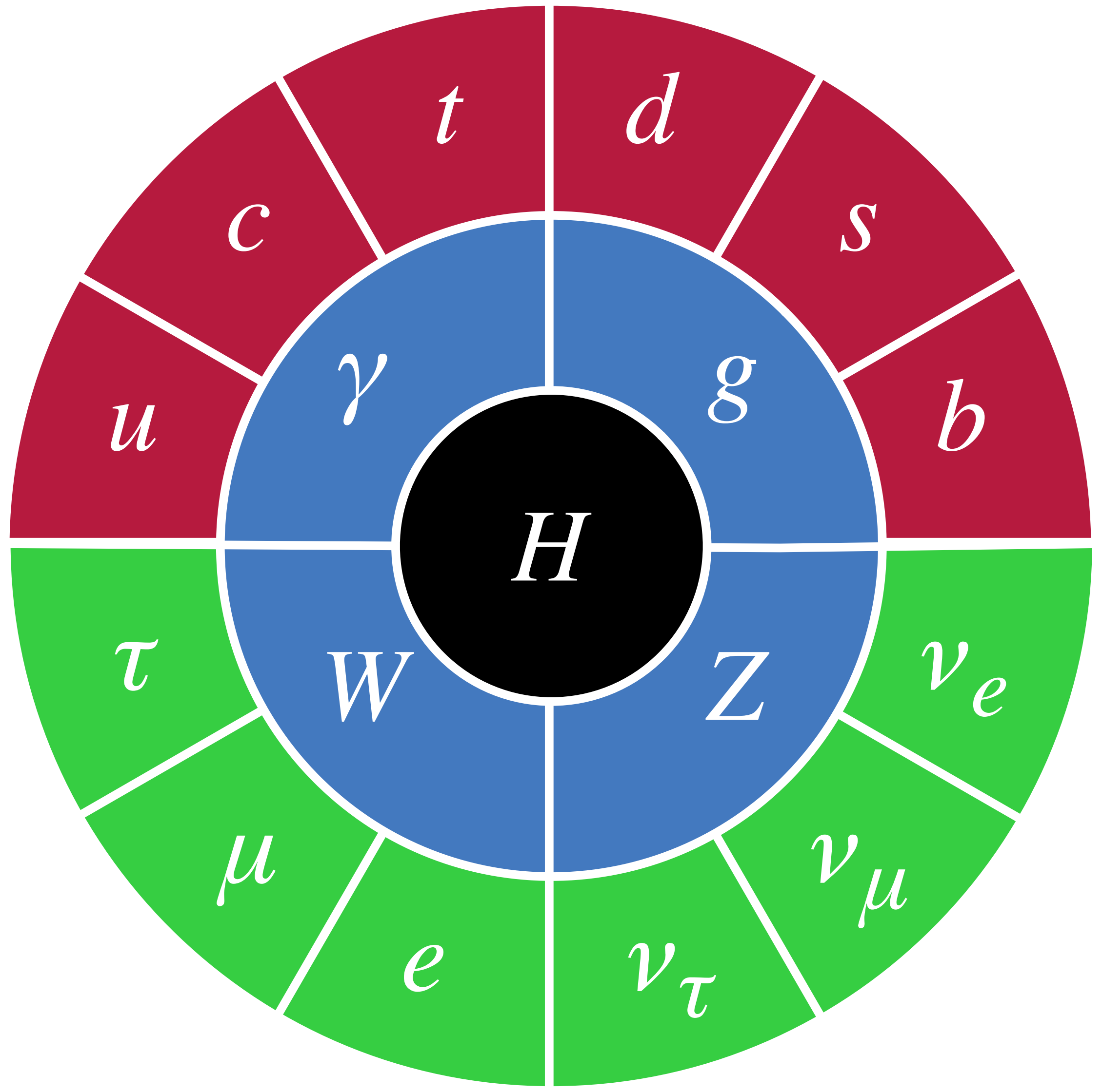


Standard Model

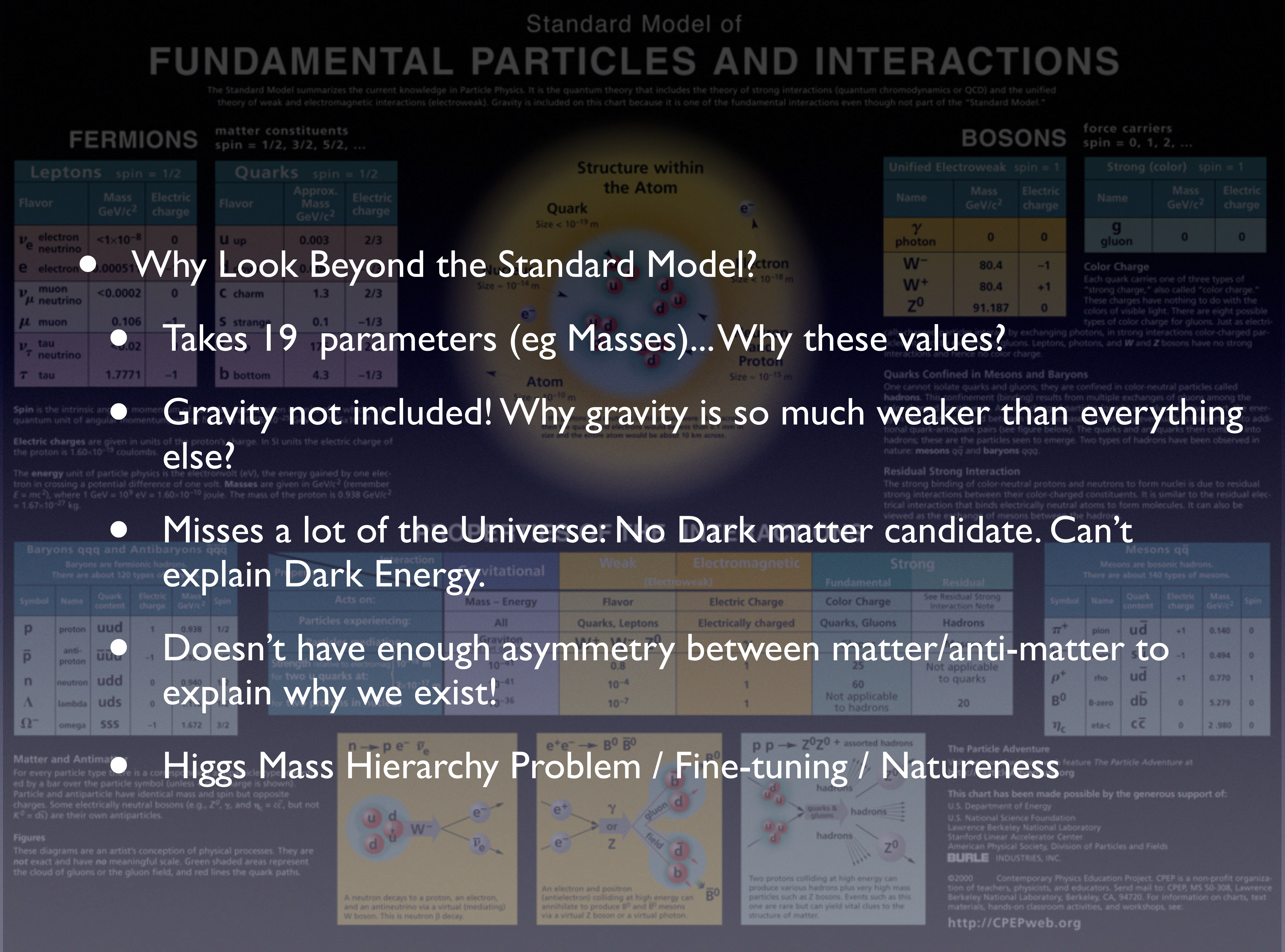
Agreement with all experimental results so far

PARTICLE PHYSICS: 19 PARAMETERS

$$
\begin{aligned}
\mathcal{L}_{SM} = \; & \underbrace{\frac{1}{4}\mathbf{W}_{\mu\nu}\cdot\mathbf{W}^{\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} - \frac{1}{4}G^{a}_{\mu\nu}G^{\mu\nu}_{a}}_{\text{kinetic energies and self-interactions of the gauge bosons}} \\
& + \underbrace{\bar{L}\gamma^{\mu}(i\partial_{\mu} - \frac{1}{2}g\tau\cdot\mathbf{W}_{\mu} - \frac{1}{2}g'YB_{\mu})L + \bar{R}\gamma^{\mu}(i\partial_{\mu} - \frac{1}{2}g'YB_{\mu})R}_{\text{kinetic energies and electroweak interactions of fermions}} \\
& + \underbrace{\frac{1}{2}\left|(i\partial_{\mu} - \frac{1}{2}g\tau\cdot\mathbf{W}_{\mu} - \frac{1}{2}g'YB_{\mu})\,\phi\right|^{2} - V(\phi)}_{W^{\pm}, Z, \gamma, \text{and Higgs masses and couplings}} \\
& + \underbrace{g''(\bar{q}\gamma^{\mu}T_{a}q)G^{a}_{\mu}}_{\text{interactions between quarks and gluons}} + \underbrace{(G_{1}\bar{L}\phi R + G_{2}\bar{L}\phi_{c}R + h.c.)}_{\text{fermion masses and couplings to Higgs}}
\end{aligned}
$$



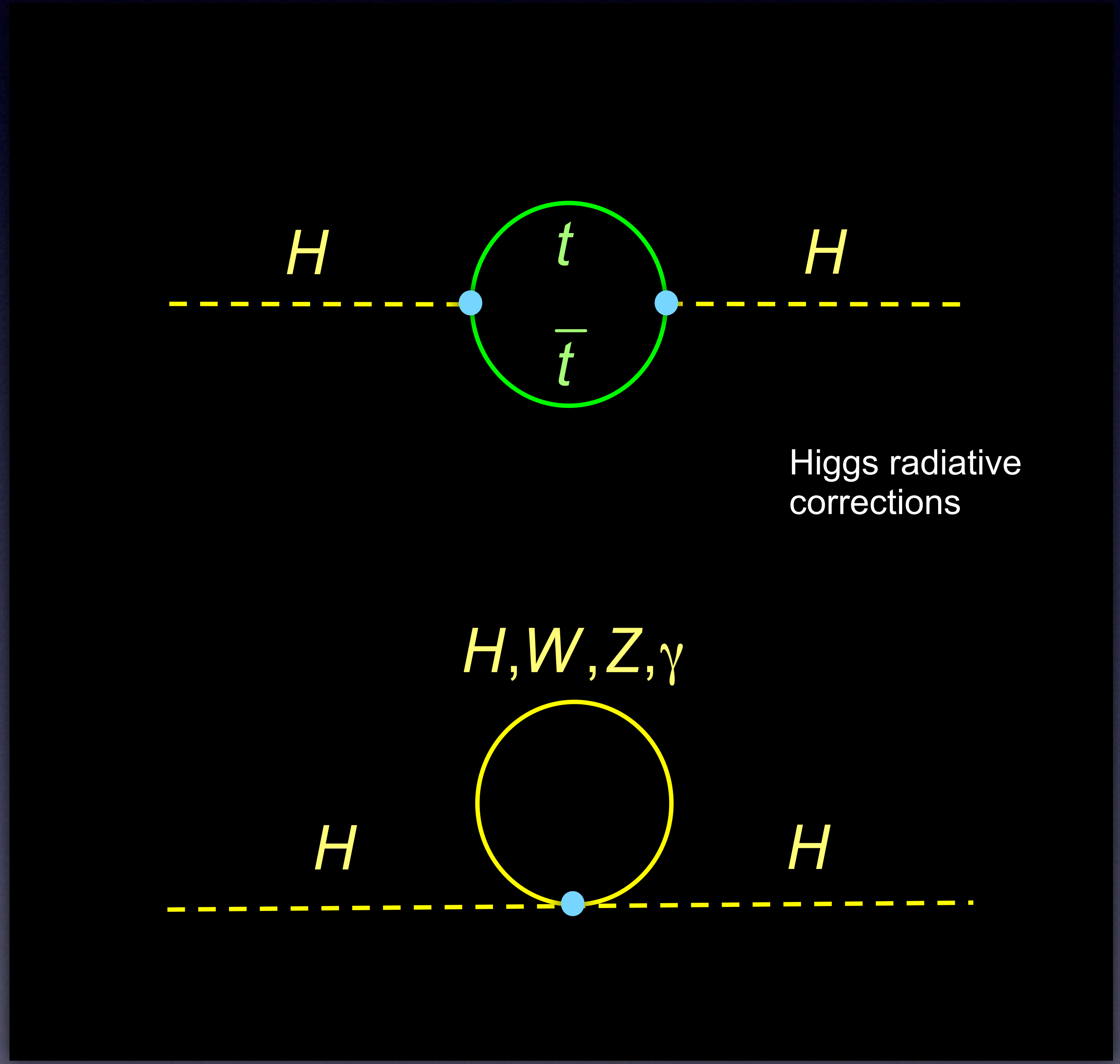
Symbol	Description	Value
m_e	Electron mass	511 keV
m_μ	Muon mass	105.7 MeV
m_τ	Tau mass	1.78 GeV
m_u	Up quark mass	1.9 MeV
m_d	Down quark mass	4.4 MeV
m_s	Strange quark mass	87 MeV
m_c	Charm quark mass	1.32 GeV
m_b	Bottom quark mass	4.24 GeV
m_t	Top quark mass	172.7 GeV
θ_{12}	CKM 12-mixing angle	13.1°
θ_{23}	CKM 23-mixing angle	2.4°
θ_{13}	CKM 13-mixing angle	0.2°
δ	CKM CP-violating Phase	0.995
g_1	U(1) gauge coupling	0.357
g_2	SU(2) gauge coupling	0.652
g_3	SU(3) gauge coupling	1.221
θ_{QCD}	QCD vacuum angle	~0
v	Higgs vacuum expectation value	246 GeV
m_H	Higgs mass	125 GeV



- Why Look Beyond the Standard Model?
  - Takes 19 parameters (eg Masses)...Why these values?
  - Gravity not included! Why gravity is so much weaker than everything else?
  - Misses a lot of the Universe: No Dark matter candidate. Can’t explain Dark Energy.
  - Doesn’t have enough asymmetry between matter/anti-matter to explain why we exist!
  - Higgs Mass Hierarchy Problem / Fine-tuning / Natureness

Why the Higgs Mass is a Problem?

- Think of the mass like a number that is constantly being "pushed around" by everything in the quantum vacuum
  - → Every particle in nature contributes a correction to every other particles's mass… usually small.
    - Heavy particles (like the top quark) give very large pushes
- Higgs is the only "scalar" particle in the SM, so corrections are different.
  - If physics continues to very high energies, → these pushes become enormous for the Higgs
- So naively, you would expect:
  - → The Higgs mass should be very large (set by the highest energy scale in nature)

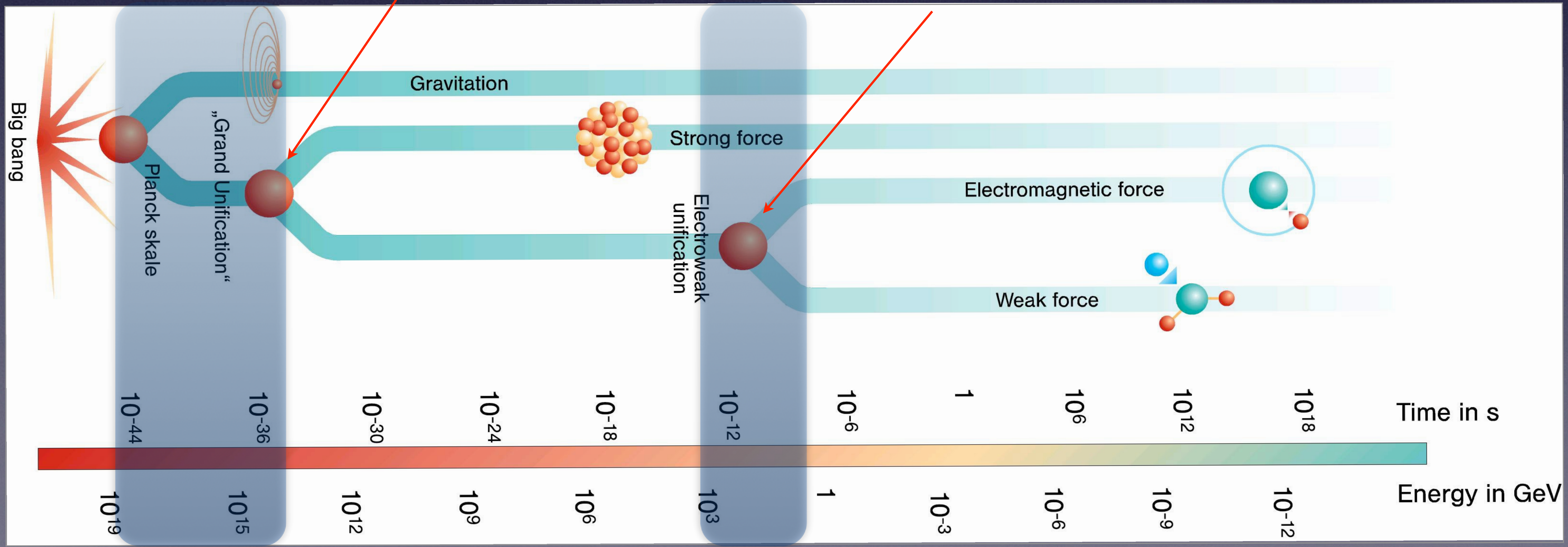


The Puzzle

- We naively expect the Higgs to be very very Heavy due to radiative corrections.
- But experimentally, the Higgs mass is only 125 GeV
  - → much smaller than, for example, the Planck scale
- To make this happen, nature would need:
  - → huge contributions that almost perfectly cancel each other
    - This is like adding 10^{16} - 10^{16} + 125 and somehow always getting exactly 125
  - That kind of precise cancellation feels unnatural

Higgs Mass w/o New Physics

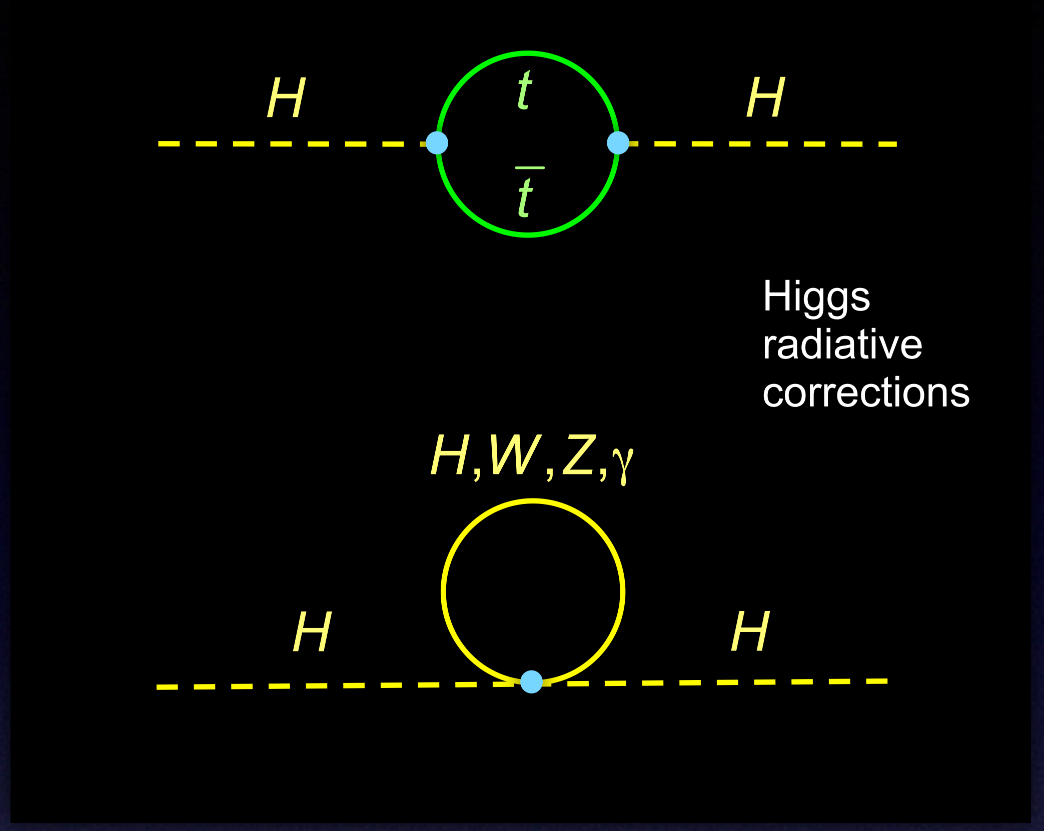
Where Higgs Mass Needs to be



Why We Expect New Physics?

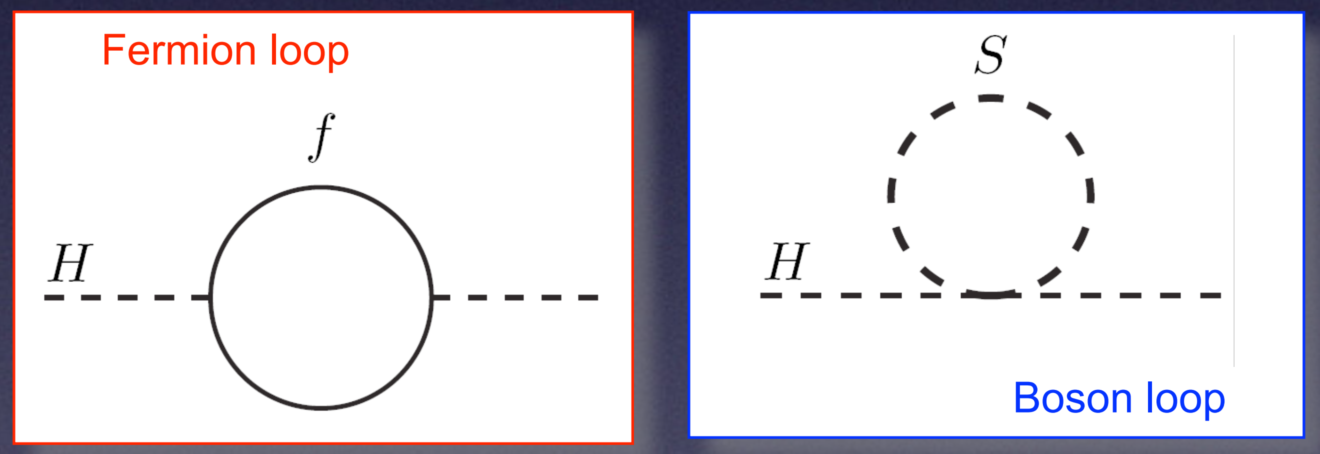
- Physicists interpret this as a sign that something is missing in the Standard Model
- The idea:
  - → New physics should automatically cancel or limit these large contributions
- Examples:
  - New particles that cancel quantum effects → Supersymmetry
  - Higgs not fundamental (composite) →Technicolor
  - Extra dimensions changing behavior at high energy → Large Extra Dimensions
- All predict new particles... usually partners to the Standard Model particles
- Some of the theories solve multiple problems:
  - Higgs Mass Fine tuning, Dark Matter, Gauge [Force] Unification

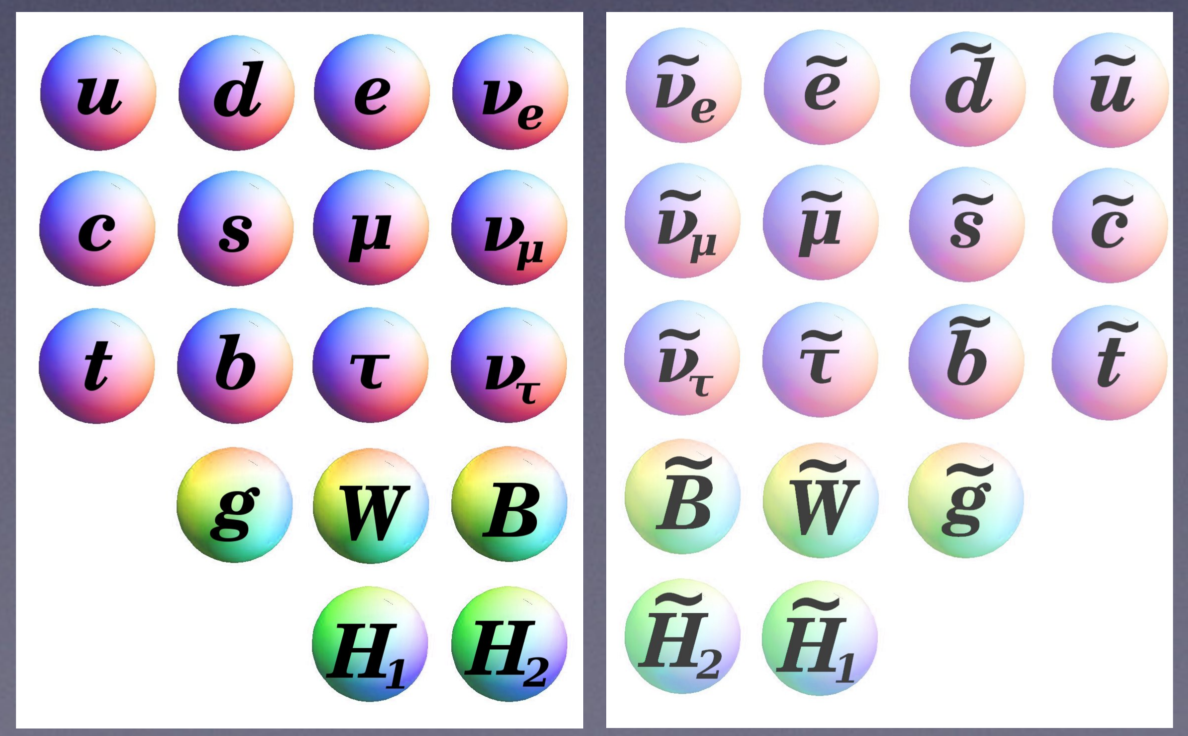
SUSY and Hierarchy Problem



- Recall the mass of the Higgs is M_{GUT} or M_{PL}, because of "infinite" radiative corrections (ie loop diagrams) that are quadratic in cut-off Λ.

- What if for every such diagram, there existed another diagram, with opposite sign?
  - Bosons/Fermions come with different signs
  - A new space-time symmetry (extension of Poincare Group) results in a new boson for every fermion and visa-versa
  - particle → sparticle, gauge boson → gaugino
- No scalar electron partner ⇒ SUSY broken
  - Still keeps the Higgs mass low



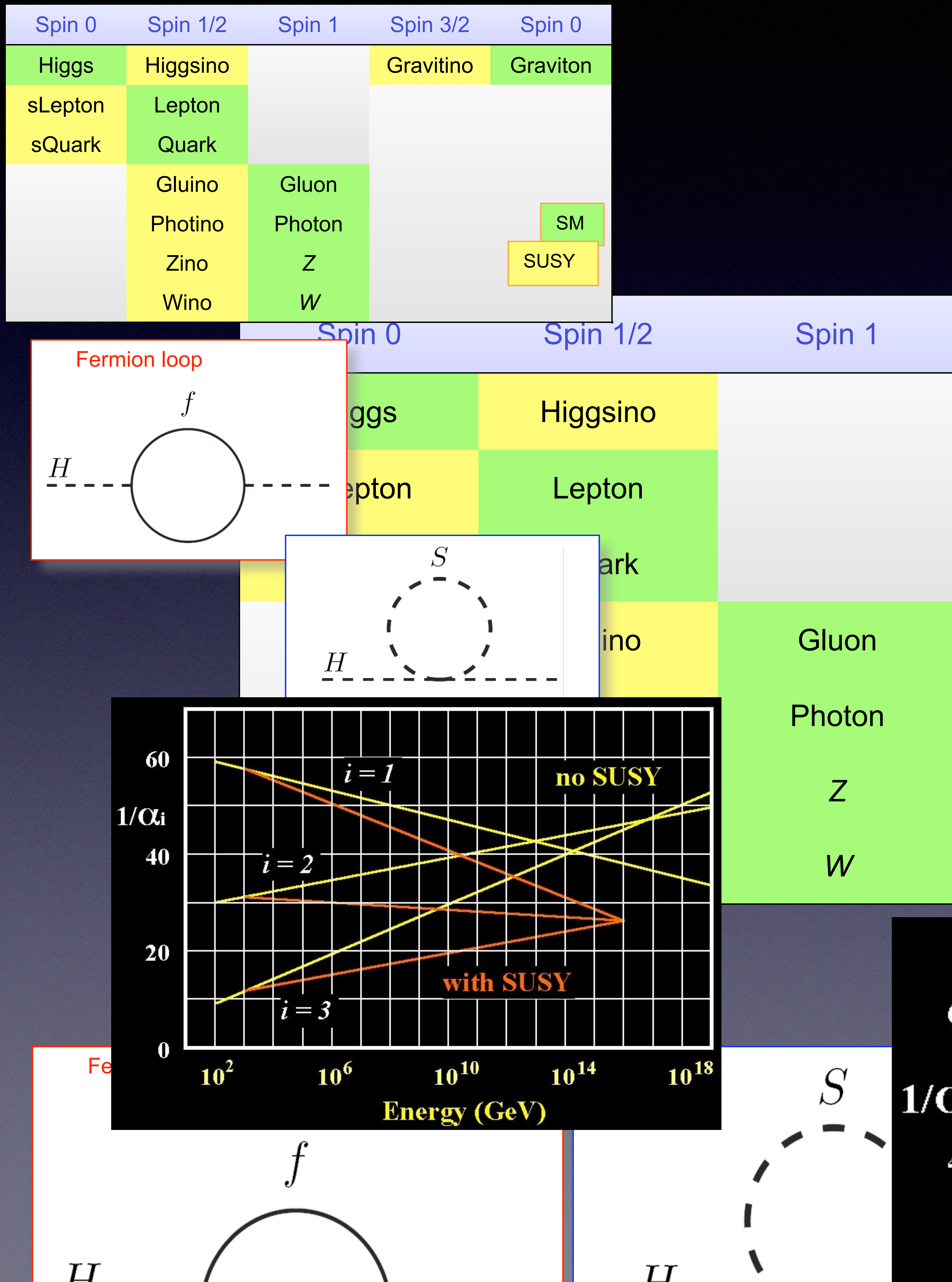


SUSY Motivation

- Aesthetic: new space-time symmetry
  - Leads to new partners for every SM particle.
  - Removes quadratic divergences (Higgs mass).

➡ Resolves Hierarchy problem

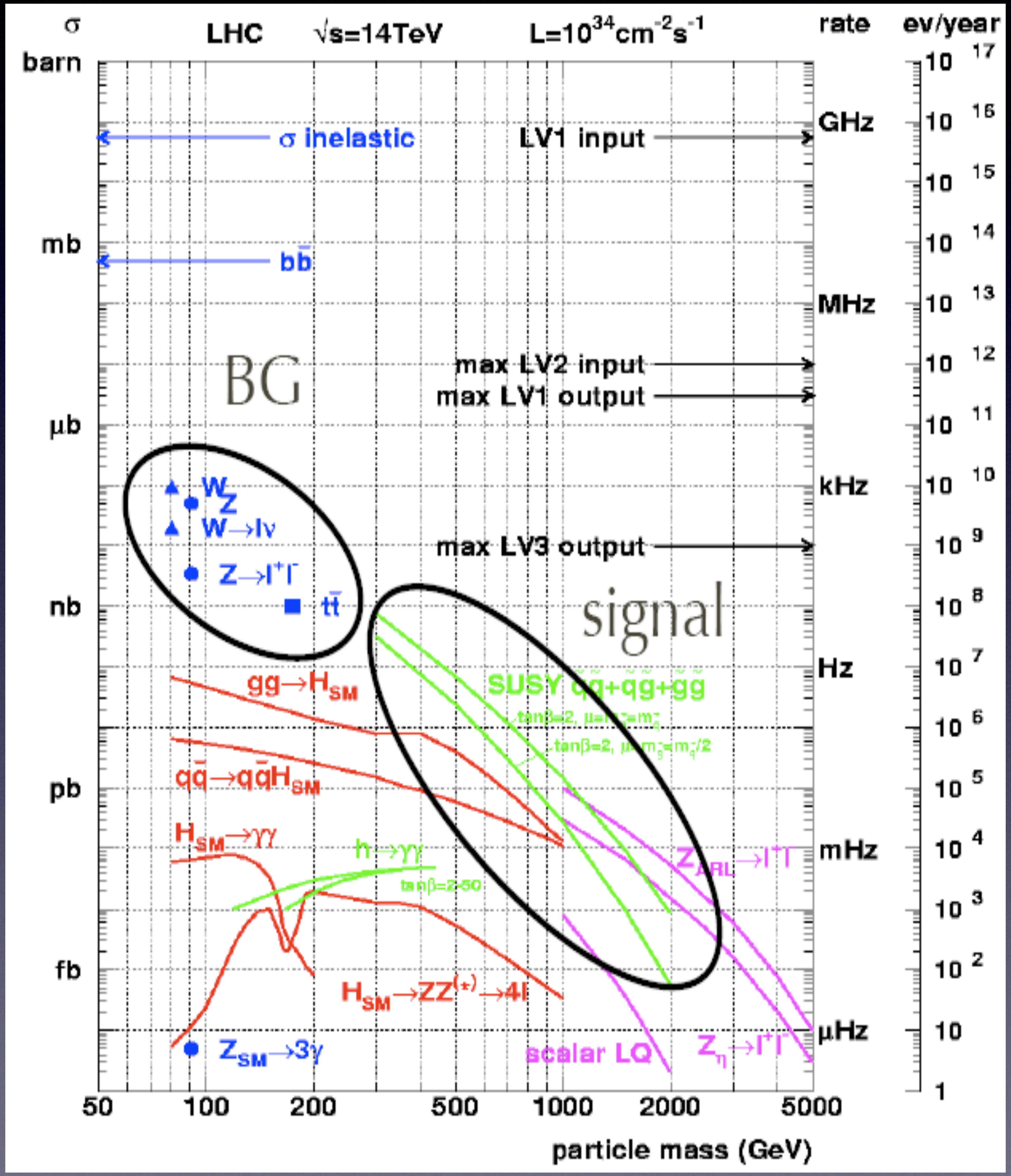
- Gauge unification
- Has Graviton
- Dark matter Candidate: The Lightest SUSY Particle can be a heavy stable neutral particle

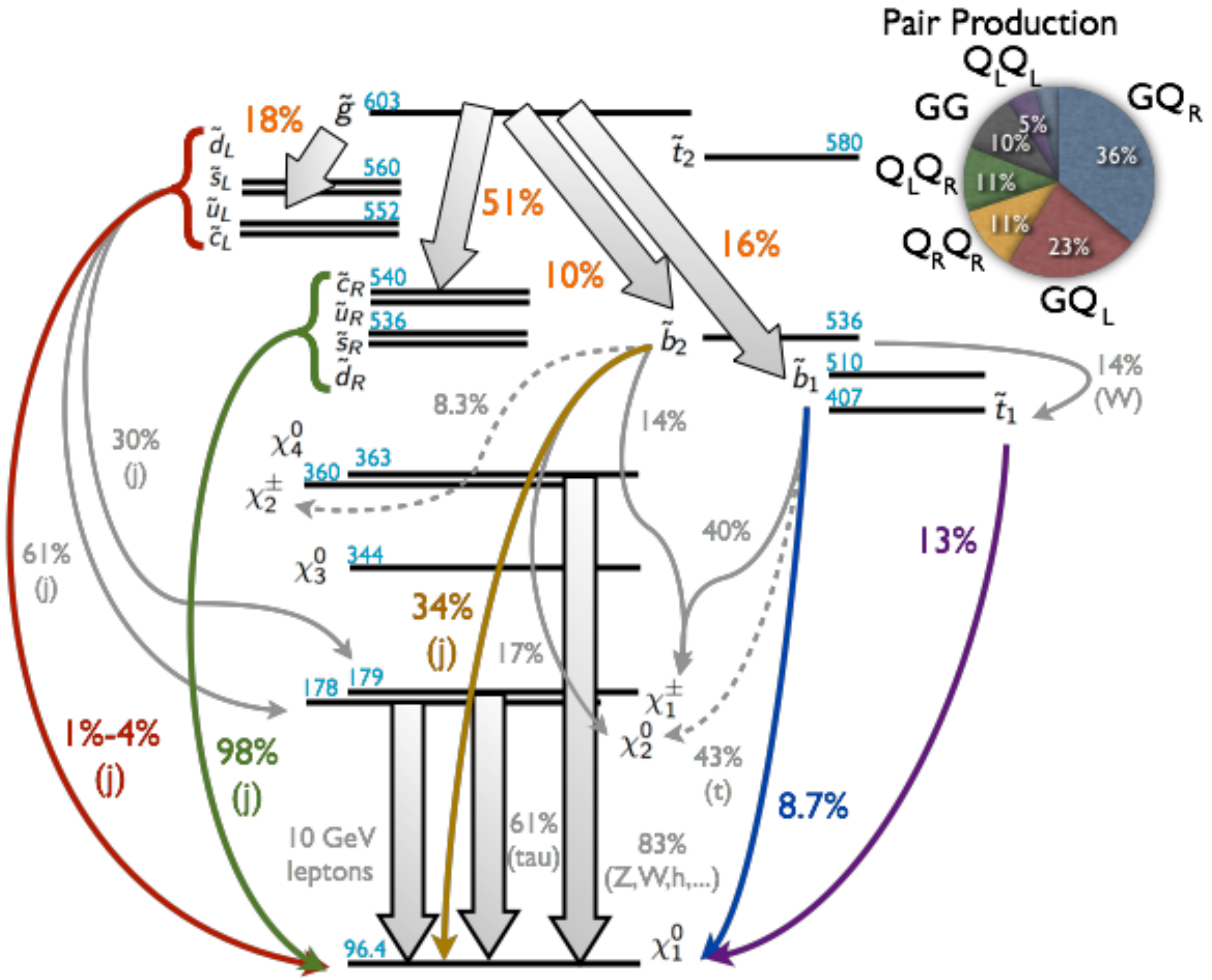


Looking for SUSY

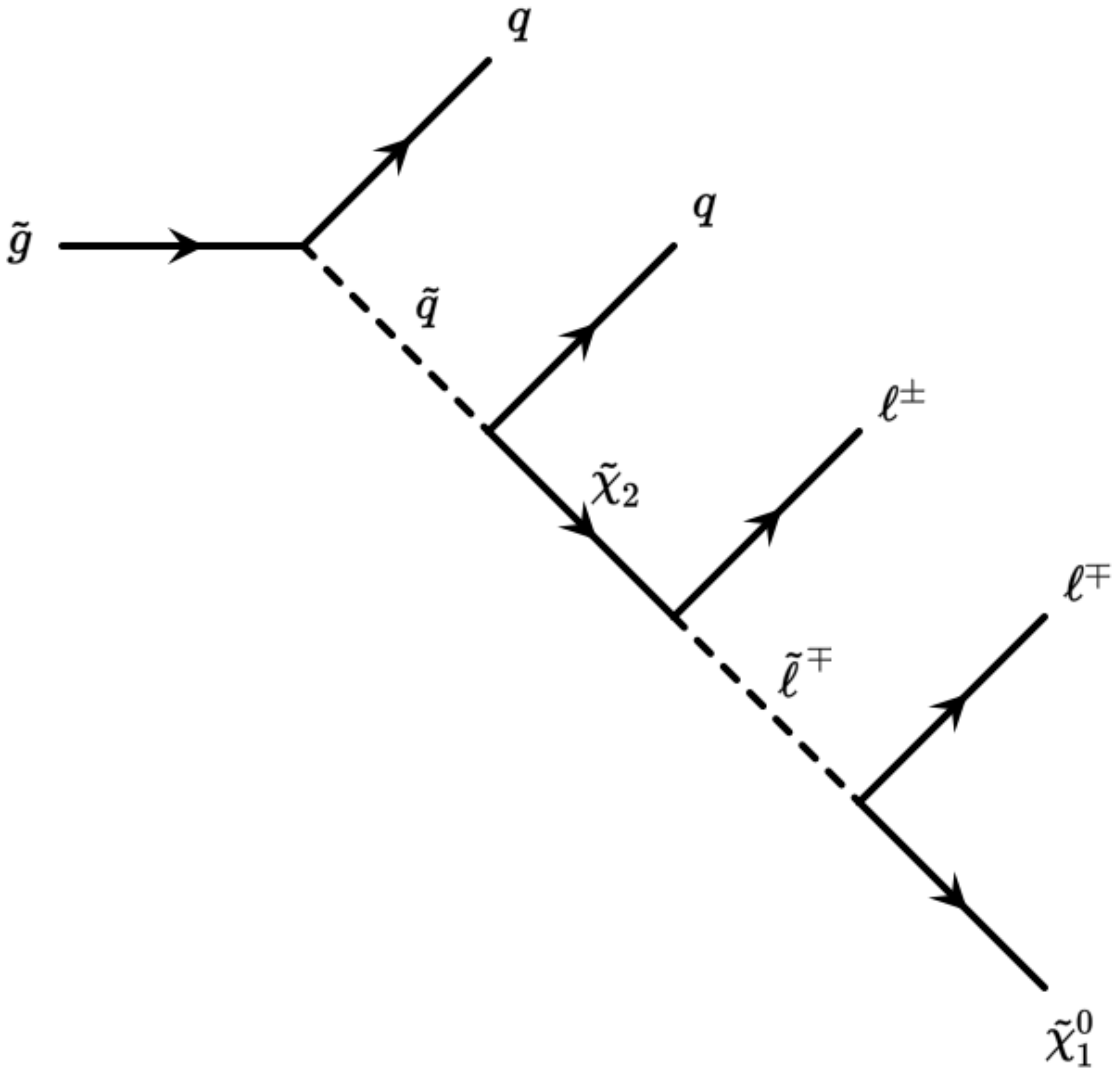
Signal vs Background

- 40 Million collisions a second
  - 50 interactions per collision
- Interesting stuff (signal) is ~ 1 in a trillion
  - The rest is background

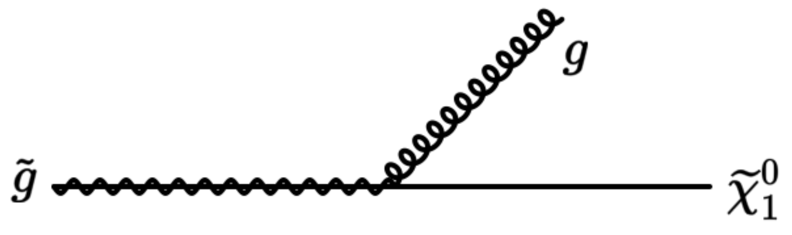



Pair Production
GG
QLQL
GQR
QLQR
QRQR
GQL
5%
10%
36%
11%
11%
23%
18%
51%
10%
16%
603
580
560
552
540
536
536
510
407
363
360
344
179
178
96.4
8.3%
14%
14%
(W)
30%
(j)
61%
(j)
40%
13%
34%
(j)
17%
1%-4%
(j)
98%
(j)
43%
(t)
8.7%
10 GeV
leptons
61%
(tau)
83%
(Z,W,h,...)

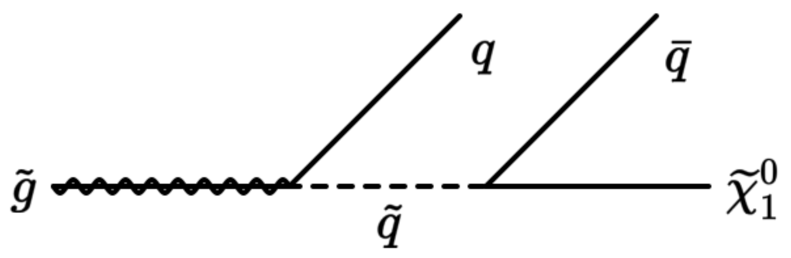
(a)


$\tilde{g}$
q
q
$\tilde{q}$
$\tilde{\chi}_2$
$\ell^{\pm}$
$\tilde{\ell}^{\mp}$
$\ell^{\mp}$
$\tilde{\chi}_1^0$

(b)


$\tilde{g}$
g
$\tilde{\chi}_1^0$

(a)


$\tilde{g}$
q
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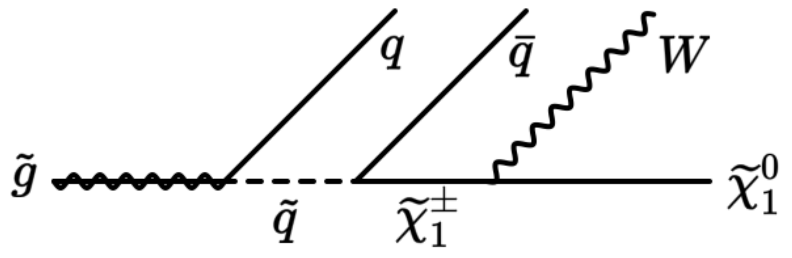
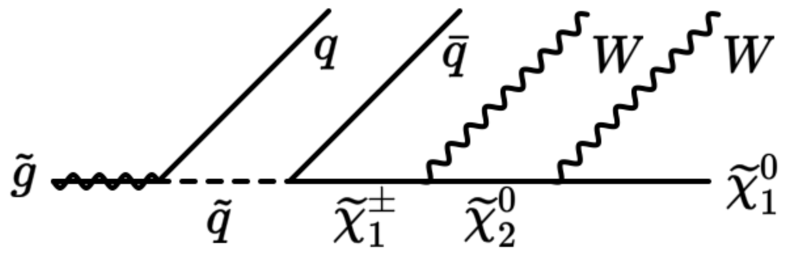
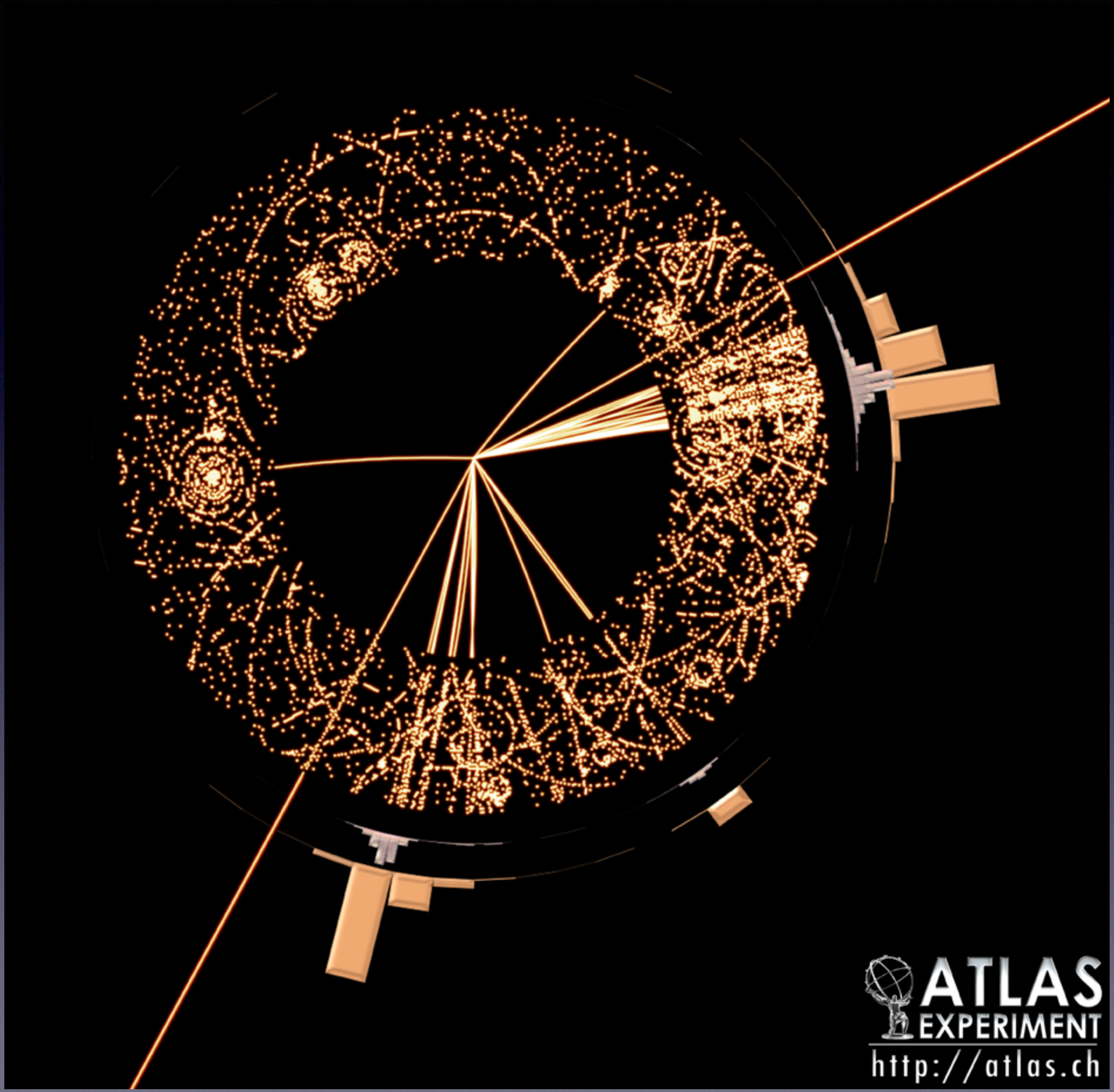
(b)

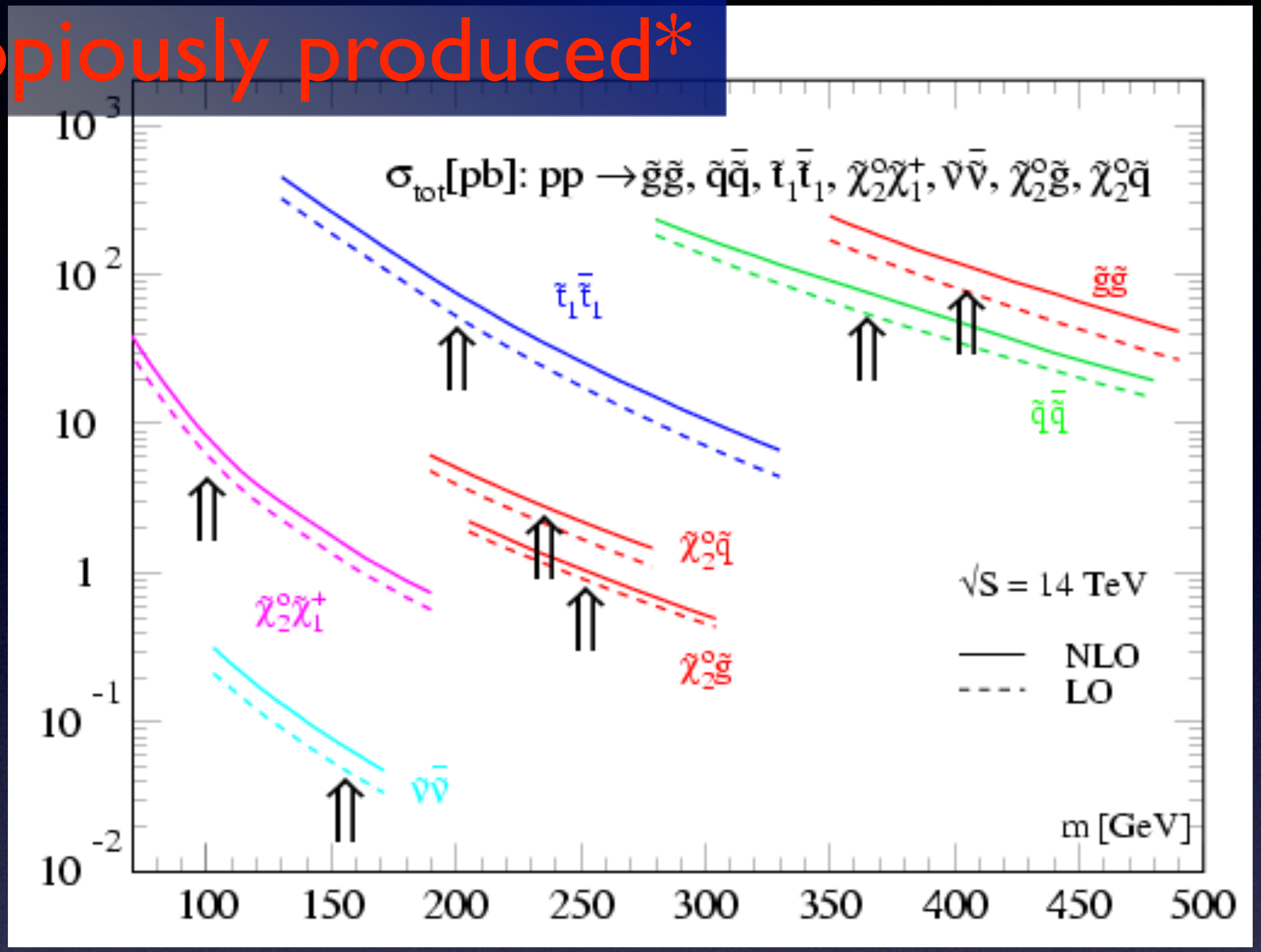

$\tilde{g}$
q
$\bar{q}$
W
$\tilde{q}$
$\tilde{\chi}_1^{\pm}$
$\tilde{\chi}_1^0$


$\tilde{g}$
q
$\bar{q}$
W
W
$\tilde{q}$
$\tilde{\chi}_1^{\pm}$
$\tilde{\chi}_2^0$
$\tilde{\chi}_1^0$

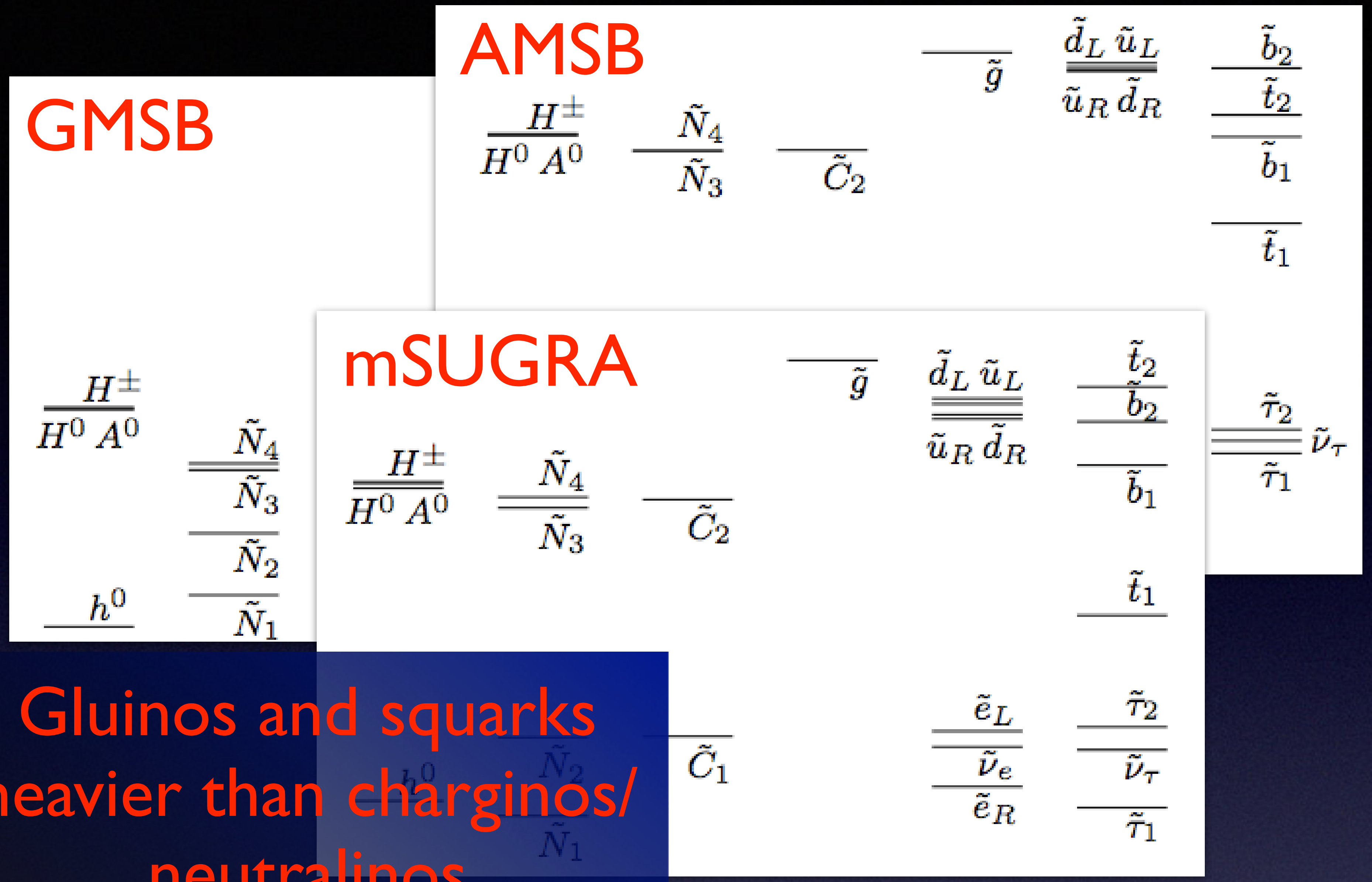

ATLAS
EXPERIMENT
http://atlas.ch

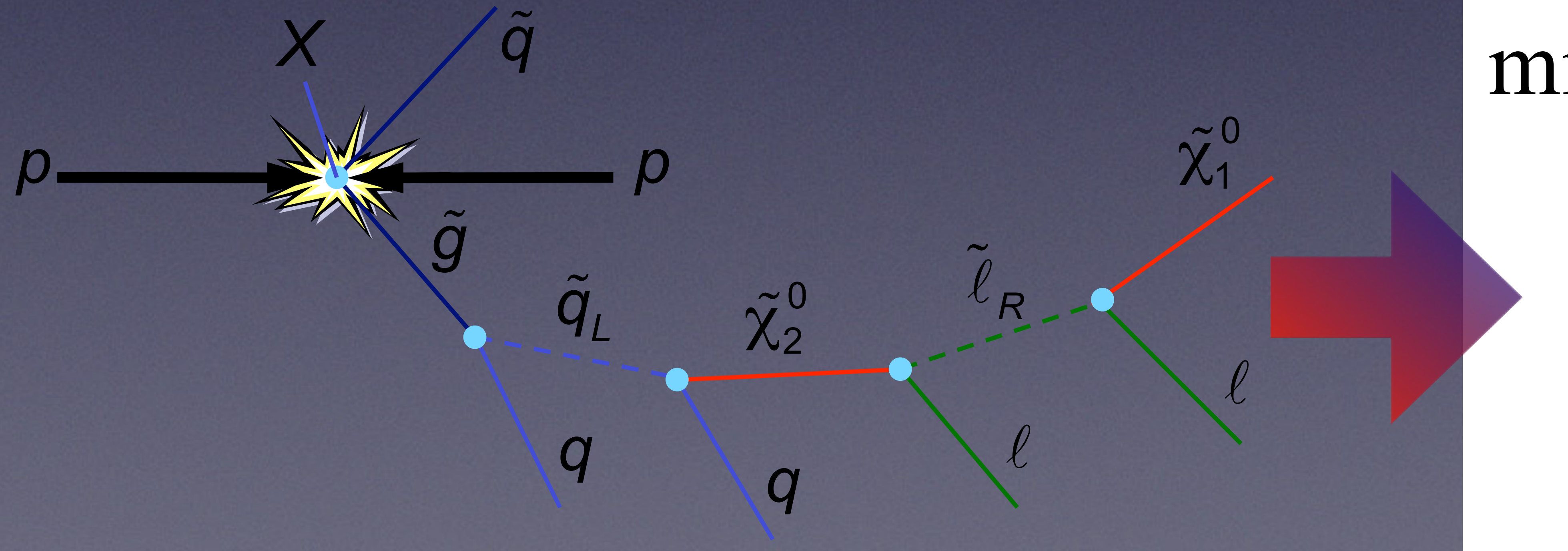
SUSY at LHC

Gluinos and squarks copiously produced*

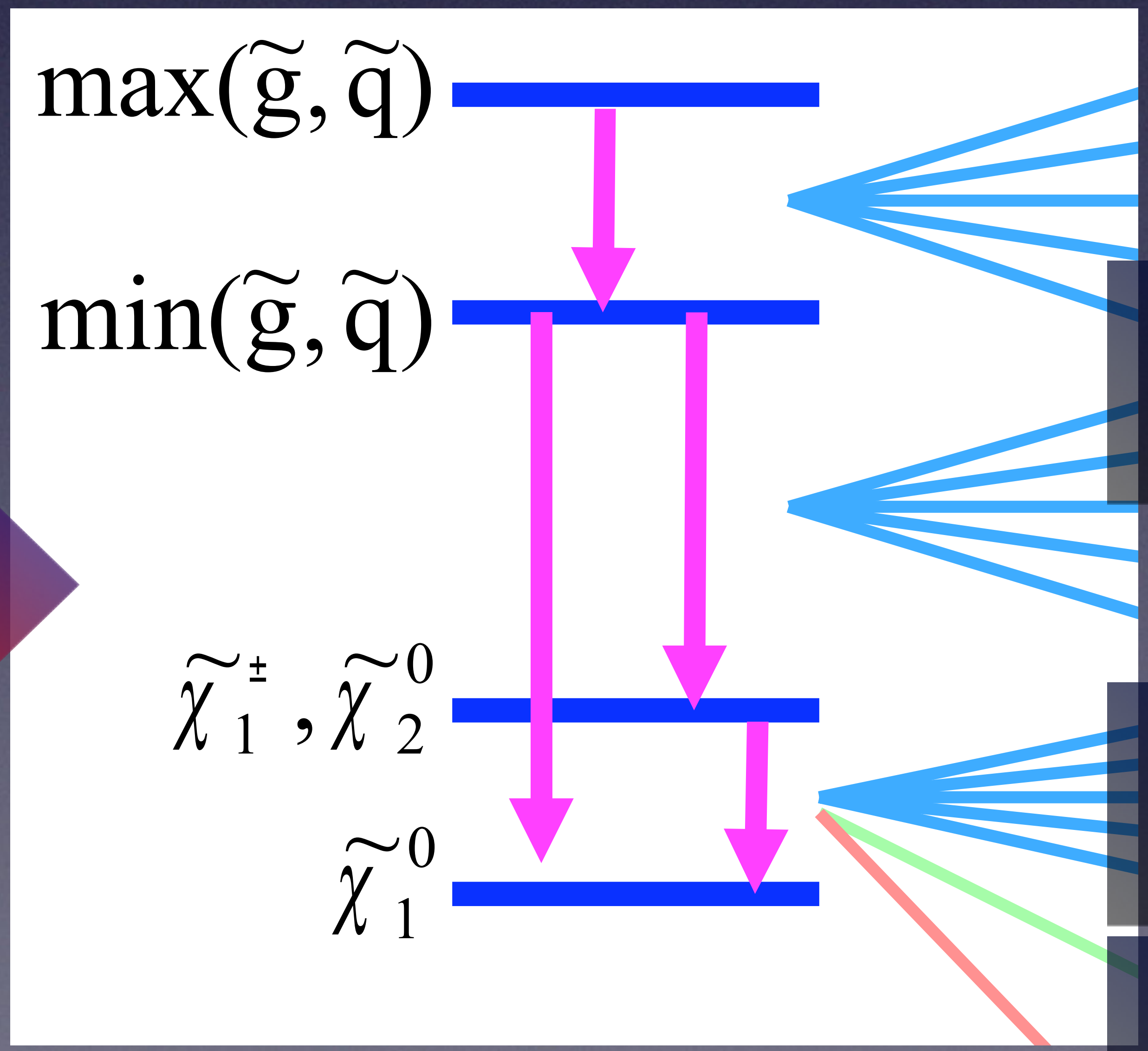


Gluinos and squarks heavier than charginos/ neutralinos





Long Cascades



Many Jets (100's of GeV)

Leptons (10's of GeV)

Missing Energy (100's of GeV)

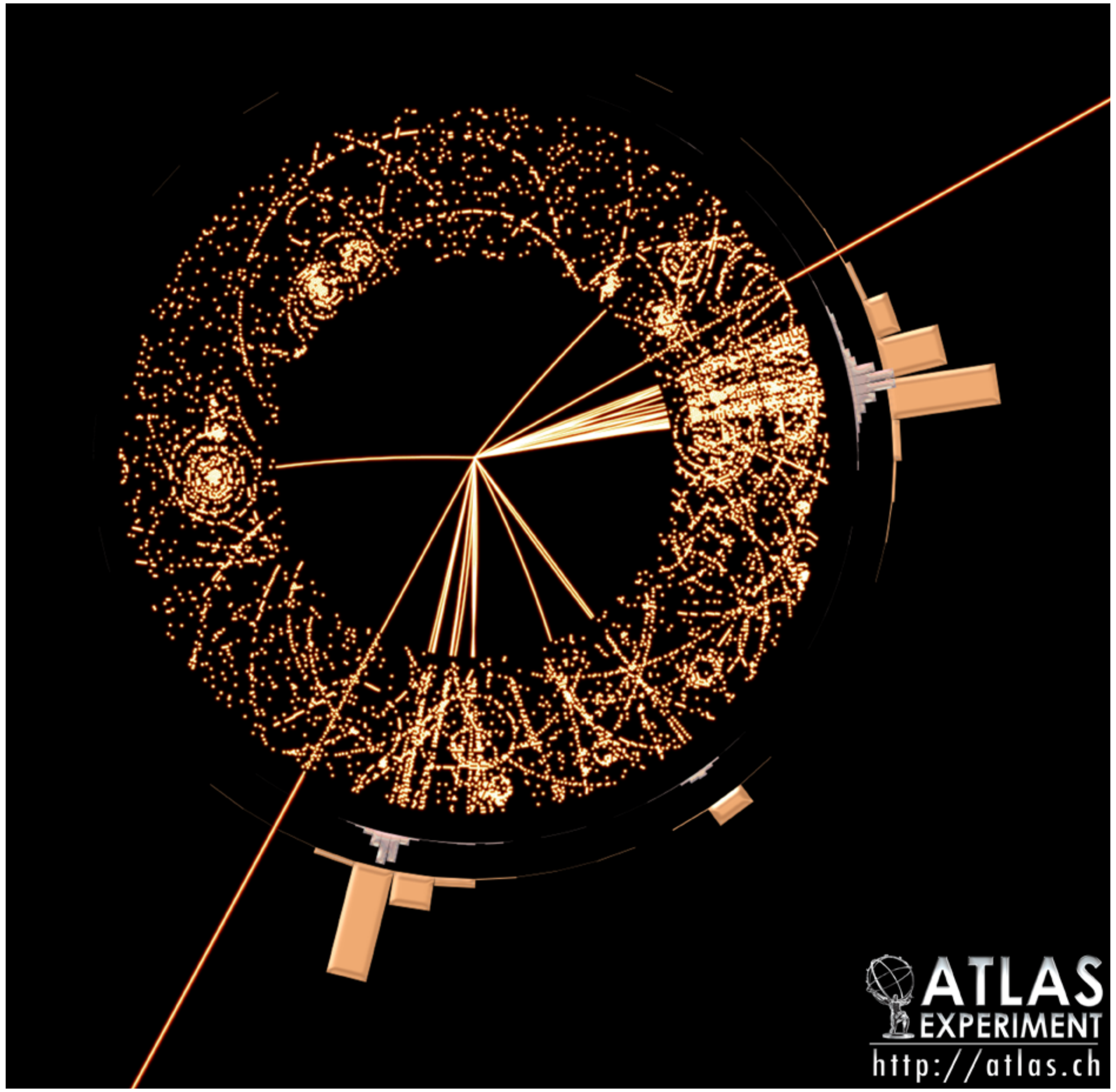
Inclusive Signatures

Signature	Motivating Model(s)	Comments
1 Jet + 0 Lepton + MET	• Large Extra Dim (ExoGraviton) • strong qG production, G propagate in extra Dim • Planck Scale is MD in 4+δ dim • Normal Gravity >> R • SUSY • qg→ISR + 2 Neutralino or squark + Neutralino	• Not primary discovery channel for SUGRA, GMSB, AMSB... but helps in characterization • Possible leading discovery for neutralino NLSP with nearly degenerate gluino
2,3,4 [b]-Jet + 0 Lepton + MET	• Squark/gluino production • squark→q+LSP, gluino→q+squark+LSP	• Possible leading squark/gluino discovery channel • Must manage QCD bkg
2,3,4 [b]-Jet + 1 Lepton + MET	• squark/gluino production with cascades which include electroweak (or partner) decays • high tan β leads to more b/t/τ's	• Lepton requirement suppresses QCD • τ's partially covered by e/μ
2 lepton + MET	• Same sign: gluino cascade can have either sign lepton... squark/gluino prod can produce same sign. • Opposite sign: squark/gluino decay mediated by Z (or partner) • Same flavor: 2 leptons from same sparticle cascade must be same flavor	• Reduced SM backgrounds for same sign • Opposite Sign-Flavor Subtraction
3 lepton + MET	• SUSY events ending in Chargino/neutralino pair decays • Weak Chargino/Neutralino production • Exotic sources	• Low SM bkgs
2 photon + MET	• GMSB models with gravitino LSP and neutralino or stau NLSP • UED- each KK partons cascade to LKP which decays to graviton + γ	• No SUSY limit (not sensitive at the time)

SUSY/Higgs Data

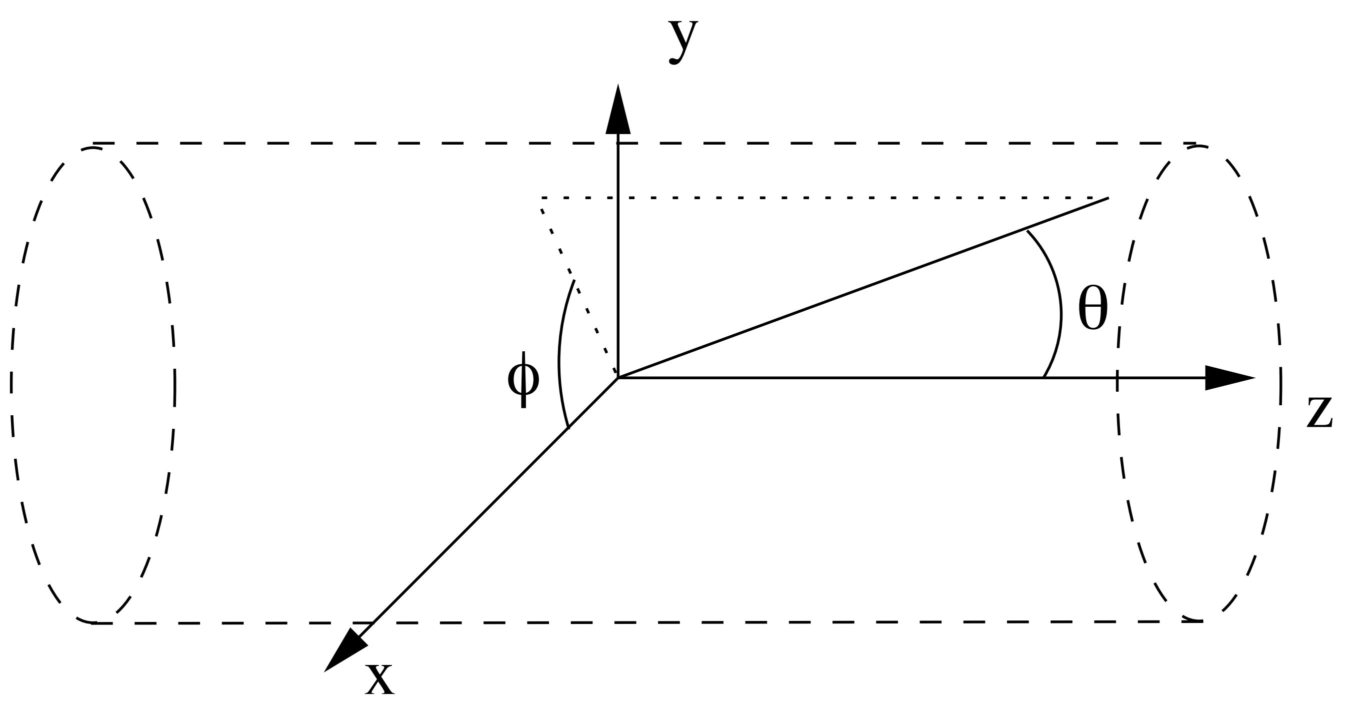
LHC Event

- The trigger LHC data is reconstructed into particle candidates.
    - Reconstruction algorithms try to find, identify, and measure particles
    - Typically there is one set of algorithms for each type of particle (corresponding to unique way the particle interacts with detector). For example:
        - *electron / photon*: Look tight deposits of energy in Electromagnetic Calorimeter. If a charged track points to it → electron otherwise photon.
        - *muon*: find track segment in inner detector and outer muon tracker.
        - *jets*: Look for broad deposits of energy in the Hadronic Calorimeter.
    - Not exclusive: Most electrons get also reconstructed as jet.
        - A process of Particle Identification and Overlap removal attempts to associate energy deposits in the detector with one unique hypothesis (as best as possible).
- Therefore the data and simulated data for an analysis will be process up to a level where every event consists of:
    - List of particle candidates, each with their momenta: electrons, photons, muons, electrons, jets (+ b-tagging), tau-jets, …
    - Event level quantities: missing energy



LHC Events

- Candidate variables:
  - For each candidate, we measure it's 3-vector in a coordinate system best suited for Hadron Colliders.
    - Reflects the fact that not all of energy of beam goes into collision → can only apply energy conservation transverse to the beam.
    - $(p_x, p_y, p_z) \rightarrow (p_T, \eta, \phi)$
  - Once you choose what to call a particle, then you can assume a mass and turn your 3-vector → 4-vector
- Event Variables: Missing transverse energy (magnitude and direction in transverse plane (ϕ))
- In this context, by raw features we mean these 4-vector and event variables.



$$p_T^2 \equiv p_x^2 + p_y^2$$
$$\vec{p}_T \equiv \vec{p} \sin\theta$$

$$\eta \equiv \frac{1}{2} \ln \frac{1+\cos\theta}{1-\cos\theta} = -\ln\left(\tan\frac{\theta}{2}\right)$$

$$E_T^2 \equiv p_T^2 + m^2 = E^2 - p_z^2$$

Data Sample

- SUSY Data: (Selected events with only 2 leptons)
  - “signal”: The truth. 1 = signal. 0 = background.
  - 2 leptons: ”l_1_pT", “l_1_eta”,”l_1_phi", "l_2_pT", "l_2_eta", “l_2_phi"
  - Missing energy: ”MET", "MET_phi"
  - Features: ”MET_rel”, "axial_MET", "M_R", "M_TR_2", "R", "MT2", "S_R", "M_Delta_R", "dPhi_r_b", "cos_theta_r1"
- Higgs Data: (Selected events with 1 lepton and 4 jets)
  - lepton pT, lepton eta, lepton phi
  - missing energy magnitude, missing energy phi,
  - jet 1 pt, jet 1 eta, jet 1 phi, jet 1 b-tag,
  - jet 2 pt, jet 2 eta, jet 2 phi, jet 2 b-tag,
  - jet 3 pt, jet 3 eta, jet 3 phi, jet 3 b-tag,
  - jet 4 pt, jet 4 eta, jet 4 phi, jet 4 b-tag,
  - Feaures: m_jj, m_jjj, m_lv, m_jlv, m_bb, m_wbb, m_wwbb.

High Level Features

- In this context, high-level features mean observables that are calculable from the 4-vectors.
  - Generally motivated by physics.
- For example: if you are expecting your signal and/or background events to include particles that are the decay products of a heavy particle,
  - you can attempt to identify the decay products and compute the mass of the heavy particle.
    - W bosons decays to 2 jets (m_jj), 3 jets (m_jjj), lepton+neutrino (m_lv), lepton+jet+neutrino (m_jlv), or 2 b-jets (m_bb)
    - Top quark decays into W + 2 b-jets (m_wbb)
  - These are an attempt to *exclusively* reconstruct W and top particles.
    - Exclusive: look for a specific particle by pull together *all* of the decay particles
- Another example: if you expect your signal and/or background events to consist of various types of particles with common traits,
  - you can attempt to construct features that capture the trait:
    - SUSY particles are heavy, therefore sum of the momenta of their decay products is large.

$$H_T(n) \equiv \sum_{i=1}^{n} p_{ji,T} \qquad M_{eff} \equiv H_T(4) + \sum_{k=1}^{m} p_{\ell k,T} + E_T^{miss}$$

    - In SUSY, the last particle in the decay chain is not detected → large missing energy that isn't a fluctuation (e.g. same direction as a jet).
    - particles in SUSY events are more broadly distributed → "spherical" rather than "jetty" events.